AGT 228
INTRODUCTION TO ANIMAL HEALTH

Learning objectives

OBJECTIVES
Understand the classification of animal diseases.
Identify sick and healthy animals.
Understand the procedures in post-mortem examination.
Identify common disease caused by bacteria, viruses, protozoa, and nutritional disorder
Understand the life cycle and symptoms of helminthes and ectoparasites.
Identify the general prevention and control of diseases in animals

What is disease?

Disease is defined as a deviation from the normal functional, structural, or behavioral status of an animal. It is not merely the absence of health but represents a significant change that can be major or minor, qualitative or quantitative, depending on the severity of the condition.
The deviation from the normal and is revealed by changes in the animal. Any animal that has a disease will show some abnormality. This change in the animal can be observed from the behavior, structure or function of the animal in question. A sick animal will look dull and weak (lethargy), stay on its own and will refuse to feed (anorexia). A dairy animal that produces milk will have a drop in the quantity of milk produced. You can also notice a change in shape in areas of the body structure of a sick animal depending on the part or organ of the body that is affected.

A comprehensive understanding of disease involves examining its causes, classifications, and the clinical signs used to recognize it.

When an animal falls ill, the observable physical or behavioral changes are referred to as clinical signs or symptoms. While some symptoms are unique to specific illnesses, others—such as lethargy, fever (pyrexia), and a loss of appetite—are common across a wide range of conditions.

These clinical signs serve as the foundation for diagnosis. Diagnosis is the methodical process of identifying the nature and cause of a disease while distinguishing it from others. To reach an accurate diagnosis, one must gather comprehensive information about the animal, as these details constitute the essential components of the diagnostic process.

The process of diagnosis include the following:

History Taking: This is all about getting the "backstory." You'll want to collect the basic details—like the animal's age, breed, sex, and name or tag number. It's also super helpful to note exactly when you first noticed them acting differently and how long they've been feeling sick.

Physical Examination: Next, it's time for a "check-up." You'll observe the animal closely for anything that looks out of the ordinary, and check their vital signs like temperature, pulse, and breathing rate. If you're on a farm, don't forget to take a quick look at their living space—like their housing conditions and what they've been eating—since the environment can often hold clues too.

Laboratory Examination: Sometimes you need a closer look that you can't get with the naked eye. This involves taking samples—such as blood, waste, skin, or even food—and sending them to a lab. Scientists then run tests to hunt for any germs or issues that might be hiding. If an animal has sadly passed away, examining their tissues can also provide vital information. These lab results are incredibly helpful for getting to the bottom of the problem and reaching a clear, accurate diagnosis.

Definitions of some terms:

Infection: The invasion and multiplication of disease-causing agents (like bacteria, viruses, or parasites) within an animal's body, which may or may not cause visible illness. Salmonellosis in poultry, caused by Salmonella bacteria multiplying in the gut

Endemic: A disease that is constantly present within a specific geographic area or population at a predictable, baseline level. Trypanosomiasis (Nagana) is endemic to many parts of sub-Saharan Africa, where tsetse flies carry the parasite

Epidemic: A sudden, widespread increase in the number of cases of a disease in a population, significantly exceeding what is normally expected in that area. Newcastle Disease spreading rapidly through a specific village's poultry flock

Pandemic: An epidemic that has spread over a very large area, such as multiple countries or entire continents, affecting a large proportion of the population. Highly Pathogenic Avian Influenza (H5N1), which has spread globally across multiple continents

Sporadic: A disease that occurs only occasionally, in scattered or isolated instances, with no clear pattern or connection between cases. Anthrax often appears as sporadic cases in cattle when they graze in areas with contaminated soil spores

Contagious: A disease that is easily transmissible from one animal to another through direct contact or close proximity. Foot-and-Mouth Disease (FMD), which is highly contagious and spreads rapidly via saliva, breath, and movement of infected livestock

Acute: A condition characterized by a rapid onset and a relatively short, often severe course of illness. Acute Bloat in cattle, where the animal can become critically ill within hours due to gas buildup

Chronic: A condition that persists over a long period, typically developing slowly and lasting for months or even years. Johne's Disease in cattle, which progresses slowly over months or years, leading to chronic weight loss and diarrhea

Mild: A term used to describe a disease or condition that causes only slight symptoms and does not severely affect the animal's overall health or performance. Mild cases of bovine papillomavirus (warts), which may cause minor skin growths without significantly impacting the animal's overall health.

Etiology: The study or description of the specific cause (or causes) of a disease. The etiology of Rabies is the Lyssavirus, while the etiology of Coccidiosis is a protozoan parasite Eimeria.

Symptoms: The observable changes or clinical signs (such as fever, loss of appetite, or behavioral changes) that indicate an animal is sick. The common symptoms of African Swine Fever include high fever, loss of appetite, and skin hemorrhaging

Pathogenicity: The ability or capacity of a disease-causing agent to produce illness or damage in a host. The high pathogenicity of African Swine Fever virus means that even a small amount can cause severe, often fatal disease in pigs

Signs of Health in animals

It is much easier to keep your animals healthy when you know what "normal" looks like. Here are the key signs that your animal is healthy:

A Healthy Appetite: A healthy animal is interested in its food. Grazing animals will eat as usual, and penned animals will be eager for their next meal. They should also drink their regular amount of water.

Bright and Alert: Look at their eyes—they should appear bright and clear. The animal should also be alert, showing interest in new sounds or sights, and their reaction to humans should be normal for their personality (either keeping a respectful distance or coming over to say hello).

Healthy Coat and Skin: Their skin should be supple, and their coat should look shiny and well-kept. If the hair looks dull or rough, it is often the first sign that they might be feeling under the weather or missing some important vitamins and minerals.

Color of Mucous Membranes: You can check the health of their blood by looking at the inner lining of their eyes, gums, or mouth. A healthy animal will have a nice "salmon pink" color in these areas—avoid anything that looks pale or a deep, vivid red.

Normal Digestion: Pay attention to their bathroom habits. They should pass droppings at a normal frequency, and the texture should be just right—neither too runny nor too dry. Also, if you listen closely to their side, you should hear gentle rumbling sounds, which means their stomach is working properly. Their urine should also have a normal, healthy color.

Natural Habits: Animals have their own routines. For ruminants, a great sign of health is that they spend the normal amount of time chewing their cud. Generally, all healthy animals will also spend their usual amount of time resting throughout the day.

Knowing these signs helps you spot any small changes early, so you can make sure your animals stay in the best shape possible.

Categories of Disease

Diseases are generally categorized into two main groups based on their origin and ability to spread:

- **Infectious Diseases:** These are caused by **pathogenic microorganisms** (bacteria, viruses, protozoa, and fungi) that invade and multiply within the host's body. They are often **contagious**, meaning they can spread directly or indirectly between animals.
- **Non-infectious Diseases:** These are not caused by pathogens and cannot spread from one animal to another. Instead, they result from factors such as nutritional deficiencies, environmental stress, or genetics.

Some criteria for disease classification

Diseases can be classified using various criteria depending on the context:

1. **By Etiology (Causative Agent):** This is the most common method, grouping diseases as bacterial, viral, protozoan, rickettsial, fungal, parasitic, or deficiency-based.
2. **By Body System:** Diseases are grouped by the system they affect, such as cardiovascular, respiratory, or reproductive diseases.
3. **By Prevention Category:** Some frameworks use categories like **neonatal** (affecting the very young), **vector-borne** (transmitted by insects), or **soil-borne** diseases.
4. **By Species:** Classification based on the animal affected, such as avian, bovine, or equine diseases.

Common Causes of Ill Health

The classification of diseases according to their **causative agents (etiology)** is the most widely used method in animal health management. This approach groups diseases based on the specific biological or environmental factor responsible for the illness.

According to the sources, diseases are classified into the following categories based on their causes:

1. Bacterial Diseases

Bacteria are small, single-celled organisms that can multiply extremely rapidly—sometimes doubling every 20 minutes. This rapid growth explains why bacterial diseases can spread through a herd in a very short time. While some bacteria are non-pathogenic and inhabit the body normally, pathogenic varieties invade and multiply within host tissues, often producing harmful toxins.

- **Examples:** Anthrax, Mastitis, Salmonellosis, Tetanus, and Brucellosis.

2. Viral Diseases

Viruses are the smallest known living organisms and represent a highly infectious form of disease. They are particularly difficult to treat because they live and replicate **inside the host's cells**; treatments aimed at the virus can often affect the host's cells as well. Most viral diseases are managed through vaccination rather than treatment.

- **Examples:** Foot and Mouth Disease (FMD), Rinderpest, African Swine Fever, and Peste des Petits Ruminants (PPR).

3. Protozoan Diseases

Protozoa are single-celled animals (unlike bacteria, which are plants). Many of these diseases are **haemoparasitic**, meaning they affect the blood, and are frequently transmitted by insect vectors.

- **Examples:** Trypanosomosis (transmitted by tsetse flies), Babesiosis (Redwater), Coccidiosis, and Theileriosis (East Coast Fever).

4. Rickettsial Diseases

These are caused by specialized organisms that were previously thought to be intermediate between bacteria and viruses but are now often classified as bacteria (such as *Ehrlichia ruminantium*). They are typically transmitted by ticks.

- **Examples:** Heartwater (Cowdriosis), Anaplasmosis, and Infectious Keratoconjunctivitis.

5. Fungal Diseases

Fungi are plants that can be microscopic or large (like mushrooms). They spread through simple cell division or by releasing spores that can travel over wide areas.

- **Examples:** Ringworm, Aspergillosis, and Epizootic Lymphangitis.

6. Parasitic Diseases

These are caused by organisms that live on or in a host at its expense. They are broadly divided into two groups:

- **Endoparasitic (Helminths):** Internal worms, including roundworms (nematodes), flukes (trematodes), and tapeworms (cestodes).
  - *Examples:* Haemonchosis and Fasciolosis.
- **Ectoparasitic:** External parasites that live on the skin, such as ticks, mites, lice, and flies.
  - *Examples:* Mange and tick paralysis.

7. Non-Infectious Etiologies

Not all diseases are caused by living pathogens. The sources identify several non-infectious causative agents:

- **Deficiency Diseases:** Caused by a lack of essential nutrients, vitamins, or minerals in the diet.
    - *Examples:* Vitamin deficiencies and Iron or Sodium deficiencies.
- **Metabolic Disorders:** Result from a disturbance in the animal's internal metabolic processes rather than a simple lack of food.
    - *Examples:* Bloat (Tympany), Pregnancy Toxaemia (Ketosis), and Milk Fever.
- **Toxicoses (Poisoning):** Caused by the ingestion of toxic plants, chemicals (like pesticides), or wrongly constituted medications.
    - *Example:* Nitrate poisoning.
- **Genetics/Hereditary Conditions:** Diseases resulting from inherited traits that cause cells to degenerate.
    - *Example:* "Dwarfing" in calves.
- **Injuries:** Physical trauma can impair function or lead to secondary bacterial infections.

Recognition and Symptoms of Disease

The recognition of disease in animals is based on identifying **deviations from their normal functional, structural, or behavioral status**. Every animal with a disease will show some form of abnormality, effective recognition requires a manager to be intimately familiar with the normal "vital signs" and natural habits of their stock so that ill-health is easily recognizable,.

Indicators of a Healthy Animal

To recognize sickness, one must first understand the signs of a healthy animal:

- **Appetite:** A healthy animal is interested in food, grazes normally, and drinks its typical amount of water,.
- **Alertness:** It appears bright and alert, responding naturally to unusual noises, sights, or the approach of humans.
- **Physical State:** The coat and skin are supple and glossy; the **mucous membranes** (found in the eyes, gums, and mouth) should be a healthy **salmon pink** color,.
- **Natural Functions:** A healthy animal passes normal droppings easily, has audible digestive rumbling, and, if a ruminant, spends several hours a day **chewing the cud**,.

General Symptoms and Clinical Signs

Symptoms, or clinical signs, are the observable changes noticed when an animal is sick. While some signs are specific to certain illnesses, several general symptoms are common to most diseases:

- **Anorexia (Inappetance):** The animal "goes off" its feed or completely refuses to eat.

- **Fever (Pyrexia):** An increase in body temperature above the normal range is a classical sign of acute infectious disease,.
- **Lethargy and Depression:** The animal looks weak, dull, or dejected and often responds poorly to external stimuli,,.
- **Behavioral Isolation:** Sick animals frequently **separate themselves from the herd**, lag behind during grazing, or seek shade while others are active.
- **Physical Changes:** These include an **abnormal posture**, an awkward gait, or a coat that loses its gloss and appears **rough or dull**,.
- **Production Drops:** In dairy animals, a sudden drop in milk quantity or a change in the milk's appearance or smell is a strong indicator of a disorder,.
- **Abnormal Discharges:** Symptoms may include heavy fecal contamination (scouring) around the tail, bloody nasal discharge, or abnormal discharges from the eyes or genital organs.

Methods of Confirming Sickness

If a farmer suspects an animal is dull or behaving abnormally, they should perform preliminary health checks to confirm the illness by measuring vital signs at rest:

1. **Body Temperature:** Measured with a thermometer; a high temperature indicates fever, while a drop can suggest blood loss, starvation, or certain long-lasting diseases.
2. **Pulse Rate:** The heart rate typically becomes **faster during fevers** and slower or weaker in wasting diseases.
3. **Respiration Rate:** This is determined by counting chest movements per minute; heavy or labored breathing is a sign of imminent danger or specific respiratory distress,.

Symptoms of some specific disease conditions

- **Anthrax:** Characterized by high fever, staggering, hard breathing, and severe bloody diarrhea.
- **Blackquarter:** Presents as **lameness and swollen muscles**, particularly in the heavy muscles of the legs.
- **Bloat (Tympany):** Recognized by a **distended abdomen**, especially on the left side, which may feel drum-like and cause the animal to breathe rapidly in discomfort.
- **Heartwater:** Notable for neurological signs like circling, **twitching of the eyelids**, and "paddling" movements when lying down.

Early recognition is vital because illness causes individual cells to break down; starting treatment quickly can stop this cellular degeneration before it causes extensive damage to vital organs.

IMPORTANCE OF IDENTIFYING SICK ANIMALS EARLY

Being able to identify sick animals within a herd is a critical skill for any livestock manager, as it directly impacts the health of the entire group, the productivity of the farm, and the safety of human consumers.

The importance of this identification can be broken down into several key areas:

1. Enabling Prompt Treatment and Minimizing Damage

The earlier a farmer can recognize and treat a sick animal, the better the outcome. Disease causes individual cells in an animal's body to break down and die. If treatment is initiated quickly, this cellular degeneration can be halted. However, if identification is delayed, the resulting damage can be extensive, particularly if it affects vital organs.

2. Preventing Herd-Wide Outbreaks

Most significant livestock diseases are infectious and **spread rapidly** among animals living in close proximity. Identifying a sick individual early allows for necessary control measures, such as **quarantine** (physical separation) or immediate treatment, to prevent a major epidemic within the herd. This is especially important because some animals may act as **carriers**, harboring disease-causing agents and transmitting them to others even if they do not yet show obvious clinical signs themselves.

3. Maximizing Productivity and Economic Return

Healthy animals are necessary to ensure maximum performance and a high **return on investment**. Disease is a major cause of production losses, manifesting as:

- Reduced weight gain and poor feed conversion.
- Drops in milk yield.
- Poor reproductive capacity and abortions.
- Reduced work capacity in draught animals.
- Death.

Even **subclinical conditions**—where an animal is sick but does not show obvious symptoms—can cause continuous economic impact by lowering performance.

4. Facilitating Accurate Diagnosis

Recognizing that an animal is sick is the first step in **diagnosis**, which is the science of determining the nature and cause of a disease. Observing specific changes in behavior, structure, or function (clinical signs) provides the essential information needed to differentiate between diseases and select the correct treatment.

5. Supporting First Aid and Emergency Care

Effective recognition allows a farmer to decide when to seek professional veterinary assistance and when to apply **first aid**. Immediate first aid can serve to arrest blood loss, reduce pain, prevent shock, or make the animal comfortable until a veterinarian can render further help.

6. Protecting Public Health

Because many livestock diseases are **zoonotic** (can be transmitted from animals to humans), identifying sick animals is a vital public health measure. Diseases like Tuberculosis, Anthrax, and Brucellosis can be acquired by humans through contact with infected animals or their products, making early detection and isolation on the farm a first line of defense for the public.

How to identify sick animal from healthy ones

Identifying sick animals from healthy ones requires a thorough understanding of an animal's normal behavior, physical appearance, and vital signs. A disease is fundamentally a **deviation from the normal** functional, structural, or behavioral state, and recognizing these abnormalities is the first step in effective herd management.

Characteristics of Healthy Animals

A healthy animal is in a state of physical, social, and mental well-being. Its status is marked by:

- **Alertness and Behavior:** Healthy animals are **bright, alert, and responsive** to their environment and human presence. They show a natural interest in unusual noises or sights. Ruminants will spend a normal amount of time resting and **chewing the cud**.
- **Appetite and Digestion:** A primary indicator of health is an **interest in food** and the consumption of normal amounts of water. Their digestive system should produce audible rumbling noises, and they will pass a normal number of droppings that are neither too loose nor too dry.
- **Physical Appearance:** The eyes are bright, and the **coat is supple, glossy, and in good condition**. Mucous membranes (found in the eyes, gums, and mouth) should be a **salmon pink color**.
- **Vital Signs:** Healthy animals maintain constant internal body temperatures, pulse rates, and respiration rates at rest. For example, the normal temperature for cattle is approximately 39°C (101.8°F), and for sheep, it is 40°C (104°F).

Characteristics of Sick Animals

When an animal is sick, it exhibits observable changes known as **clinical signs or symptoms**:

- **Behavioral Changes:** Sick animals often become **dull, weak (lethargy), or depressed**. A classic sign of sickness in a herd is an animal **separating itself from its mates**, lagging behind during grazing, or seeking shade while others are active. They may also show a poor or slow response to external stimuli.
- **Anorexia (Inappetance):** One of the most common signs of illness is when an animal **goes off its feed** or refuses to eat.
- **Physical Abnormalities:**
    - **Posture and Movement:** The animal may adopt an **abnormal posture**, showing a dejected appearance or an awkward gait.

    - **Coat and Skin:** The coat often loses its glossy appearance, becoming **rough or dull**.
    - **Discharges:** You may notice **heavy fecal contamination** around the tail or abnormal discharges from the eyes, nose, or genital organs.
- **Altered Physiological Signs:**
    - **Fever (Pyrexia):** An increase in body temperature is a classical sign of infectious disease. Conversely, a drop in temperature can indicate starvation, blood loss, or certain long-lasting diseases.
    - **Respiration and Pulse:** Sick animals may exhibit **heavy or labored breathing** even when at rest. The pulse may become faster during fevers or slower and weaker in wasting diseases.
- **Production Losses:** In dairy animals, a sudden **drop in milk quantity** or a change in its physical appearance or smell is a strong indicator of a disorder.

Summary of Vital Signs for Recognition

Comparing an animal's current state to these standard normal ranges helps confirm illness:

Animal Type	Normal Temp (°C)	Normal Respiration (per min)	Normal Pulse (per min)
Horse	37.7 - 38.6	8 - 15	36 - 42
Cattle	38.3 - 39.0	12 - 16	45 - 60
Sheep	38.8 - 40.0	12 - 30	70 - 80
Goat	38.8 - 40.6	20 - 30	70 - 90

Note: It is important to remember that some animals may be **carriers**, harboring disease-causing agents and transmitting them to the flock without showing any obvious outward signs of disease themselves. Regular health checks and diagnostic tests are necessary for a complete assessment.

PHYSICAL EXAMINATION OF ANIMALS FOR INFECTION

A physical examination is a core component of diagnosing animal diseases and involves a systematic observation of the animal's body to detect any deviations from its normal state. By examining specific areas, a manager or veterinarian can identify clinical signs that point to particular illnesses.

Examination of the Head and Sense Organs

- **Eyes:** Healthy eyes are **bright and alert**. During an exam, check the **mucous membranes** around the eye; they should be a **salmon pink color**. Abnormalities include dullness, inflammation, or **discharges** (serous or mucopurulent), which may be caused by foreign bodies like seeds and husks or diseases like Rinderpest and PPR. In cases of Heartwater, you may notice **twitching of the eyelids**.
- **Nostrils:** The nose should be checked for any **abnormal discharges**. For example, Anthrax can cause a **bloody nasal discharge**, while diseases like Rinderpest and PPR result in mucopurulent discharges.

- **Mouth and Tongue:** Examine the gums and inside of the mouth for the normal salmon pink color. Look for **lesions, erosions, or vesicles** on the tongue and lips, which are characteristic of Foot and Mouth Disease or Rinderpest. You should also look for **petechial haemorrhages** (small red spots) under the tongue, a sign of Theileriosis. Excessive **salivation** can indicate "choke" (an esophageal obstruction) or Foot and Mouth Disease.
- **Ears:** While the sources do not detail an internal ear exam, they note that a healthy animal should be **responsive to unusual noises**, showing a natural interest in its surroundings.
- **Mandibular Region (Under the Jaw):** Check for **emphysematous swelling** or "bottle jaw" (inflammation of the mandibular region), which is a classic sign of internal parasites like Haemonchus contortus.

Examination of the Neck and Abdomen

- **Neck:** Observe the animal for **opisthotonos** (twisting of the neck), which is a neurological symptom of Heartwater. The neck area should also be checked if the animal is coughing or showing signs of "choke," as food materials can become lodged in the gullet or thoracic region.
- **Abdomen:** The left side of the abdomen should be specifically inspected for **ruminal distension**. A distended, drum-like abdomen on the left side is a primary indicator of **bloat (tympany)**. General abdominal distension can also indicate liver enlargement in diseases like Fasciolosis.

Examination of the Legs and Rear

- **Legs and Joints:** Observe the animal's **gait and posture**. **Lameness** can indicate Foot Rot, Blackquarter, or Foot and Mouth Disease. Look for specific joint issues, such as **hygroma** (inflammation of the knee), which is a symptom of Brucellosis. In Foot Rot, examine the **interdigital skin** for injuries that allow bacteria to enter.
- **Anus and Tail:** Check the mucous membrane at the entrance of the anus for the healthy salmon pink color. The area around the tail should be inspected for **heavy fecal contamination** (scouring), which suggests digestive disorders or heavy worm infestations.
- **Genital Organs:** Look for any **abnormal discharges** from the genital organs. In females, check the **vulva** for petechial haemorrhages, which can occur in diseases like Theileriosis.

General Body Condition

- **Skin and Coat:** A healthy animal has a **supple, glossy coat**. During the exam, look for a **rough or dull appearance**, which is a general sign of ill health. Inspect the skin for **crusted plaques** (Dermatophilosis), **vesicles** (Foot and Mouth Disease), or **swollen muscles** (Blackquarter).
- **Mucous Membranes:** Beyond the eyes and mouth, mucous membranes throughout the body (including the gums and anus) serve as indicators of the blood's condition and should generally be **salmon pink**. Pale membranes often indicate **anemia** caused by internal parasites or blood-borne protozoa.

IMPORTANCE OF POSTMORTEM EXAMINATION IN TREATMENT AND PREVENTION OF ANIMAL DISEASES AND DISORDERS

Post-mortem examination, or necropsy, is a foundational diagnostic tool in veterinary medicine, playing a critical role in both the treatment of existing herd infections and the long-term prevention of disease outbreaks. By determining the exact cause of death, it allows veterinarians and producers to implement targeted, evidence-based management, and therapeutic strategies, rather than relying on guesswork.

Here is a detailed breakdown of the importance of post-mortem examinations in animal health:

1. **Importance in the Diagnosis and Treatment of Diseases**

Determining the Cause of Death: A necropsy is often the quickest and most effective way to identify the specific pathogen (bacteria, virus, parasite) or environmental factor that caused an animal's death.

Confirming Clinical Diagnoses: It allows clinicians to verify whether their pre-mortem diagnosis was correct, providing valuable feedback that improves future diagnostic accuracy.

Informing Targeted Therapy: By identifying the specific disease, treatment can be precisely targeted to the surviving animals, rather than using broad-spectrum antibiotics or treatments that may be ineffective, thus reducing antimicrobial resistance.

Assessing Treatment Efficacy: It enables veterinarians to evaluate if a particular treatment or vaccination protocol used before the animal's death was actually working.

2**. Importance in the Prevention of Disease**

Preventing Future Outbreaks: Identifying a contagious disease early allows for immediate action—such as isolation, vaccination, or treatment—to protect the rest of the herd or flock.

Identifying Nutritional/Environmental Issues: Post-mortems can reveal non-infectious causes of death, such as toxicosis or nutritional imbalances (e.g., copper deficiency in sheep), prompting immediate changes in diet or husbandry.

Surveillance of Emerging Diseases: Routine post-mortems are crucial for detecting new or re-emerging diseases in animal populations.

Controlling Zoonotic Threats: Many animal diseases can be transmitted to humans. Post-mortems are vital for identifying these pathogens early (e.g., Anthrax, Avian Influenza, Rabies), thereby safeguarding public health.

3**. Economic and Practical Benefits for Producers**

Reducing Mortality Rates: By identifying the root cause of death, producers can take proactive measures to prevent similar losses in the future, improving overall farm profitability.

Evaluating Management Practices: It helps identify if farm practices, such as feeding, housing, or handling (e.g., disbudding or weaning), are contributing to high mortality.

Legal Evidence: In cases of suspected cruelty, neglect, or insurance claims, a detailed post-mortem report serves as crucial, evidence-based documentation.

Key Considerations for Effective Post-Mortem

Timing: "Fresh is best"—a post-mortem should be done as soon as possible, ideally within 24 hours, to avoid tissue decomposition, which can obscure the cause of death.

Professional Expertise: A thorough, systematic examination by a veterinarian or specialized pathologist is essential to identify subtle lesions.

Sample Collection: A, complete necropsy involves not just looking at organs, but collecting samples for histopathology, microbiology, or toxicology to confirm findings.

IDENTIFYING POSTMORTEM CHANGES IN SLAUGHTERED LIVESTOCK

Here is a guide to identifying changes in key systems and organs:

1. **Respiratory System (Lungs and Trachea)**

Normal: Soft, spongy, light pink or reddish, floats in water.

Changes to Identify:

Pneumonia: Lungs are firm, liver-like (consolidation), and discolored (dark red/purple).

Adhesions: Lungs stick to the chest wall (pleuritis).

Parasites (Hydatidosis): White, fluid-filled cysts (hydatid cysts) in lungs.

Abscesses: Pus-filled pockets (common in goats/sheep).

Emphysema: Lungs are distended, light in color, and crackle when touched (air trapped).

Anthracosis: Black pigmentation (often in urban-raised animals).

2. **Alimentary System (Liver, Intestines, Stomach)**

Liver Normal: Reddish-brown, smooth, firm but pliable.

Liver Changes:

Fascioliasis (Liver Fluke): Bile ducts thickened, pale/white, gritty, or liver is brown and enlarged.

Cirrhosis: Firm, fibrous, hardened, often shrunken (chronic infection).

Abscesses/Tumors: Abscesses (pus) or nodules (tumors).

Telangiectasis ("Plum Pudding"): Bluish-black, sunken spots (common in old cattle).

Milk Spots (Pigs): White fibrous scars (ascarid larvae migration).

Intestines/Stomach Changes:

Pimply Gut (Oesophagostomum): Small, firm nodules on the intestine wall.

Enteritis/Inflammation: Red, thickened, or bleeding gut lining.

Peritonitis: Inflammation of the intestinal lining (fecal contamination or infection).

3. **Urinary System (Kidneys)**

Normal: Dark red, smooth (except in cattle, which are lobulated).

Changes to Identify:

Nephritis: Kidney is pale, swollen, with grey or red spots (infection).

Congenital Cysts: Clear, fluid-filled sacs.

Hydronephrosis: Enlarged kidney filled with urine due to blockage.

Melanosis: Black pigment spots.

4. **Reproductive System**

Changes to Identify:

Metritis/Pyometra: Pus-filled, thickened, or inflamed uterus.

Retained Placenta: Fetal membranes still in the uterus (indicates post-calving infection).

Ovarian Cysts/Tumors: Irregular, enlarged, or nodular ovaries.

5. **Cranium (Brain/Head)**

Changes to Identify:

Coenurus cerebralis (Gid/Sturdy): Fluid-filled cysts in the brain, often causing skull softening.

Actinobacillosis (Lumpy Jaw): Swelling and abscesses in the jaw/tongue.

Abscesses: Pus-filled pockets.

6**. General Carcass Conditions (Systemic)**

Jaundice: Yellowish discoloration of fat, tissues, and cartilage.

Emaciation: "Bony" carcass, loss of fat, watery/jelly-like fat (cachexia).

Septicemia: Enlarged, wet, red lymph nodes, poor bleeding, petechial (pinpoint) hemorrhages.

Bruising/Fractures: Localized discolored areas (common on limbs).

ANALYZING FAECES, URINE AND BLOOD FOR MICROORGANISMS AND OTHER DISEASE SIGNS

1**. Analysis of Faeces (Manure Evaluation)**
Faecal analysis helps detect internal parasites, bacterial infections, and digestive issues.

Parasite Detection: Microscopic examination (e.g., fecal egg counts) identifies parasite eggs, indicating gastrointestinal worm infestations.

Visual Signs:

Diarrhea (Scours): Indicates bacterial/viral infections like Salmonella or Rotavirus.

Blood/Mucus: Suggests coccidiosis, swine dysentery, or proliferative hemorrhagic enteropathy.

Consistency: Constipation (hard, dry feces) may indicate metabolic issues or dehydration.

Rapid Tests: In-house kits can diagnose scour-causing organisms in calves.

2**. Urinalysis (Urine Analysis)**
Urinalysis is used to assess kidney function, hydration, and urinary tract infections.

Physical Inspection: Normal urine is clear and yellow/amber. Cloudiness or strong ammonia smells may indicate bacterial infection.

Discoloration: Red, pink, or brown (hematuria/hemoglobinuria) indicates blood in the urine, often caused by kidney stones, infections, or toxins.

Reagent Strips (Dipsticks): Used to check for pH, glucose, protein, and blood, which can point to metabolic disorders or kidney damage.

3. Blood analysis helps monitor overall health, infections, and metabolic deficiencies.

Hematology (Blood Count): Identifies anemia, high parasite loads, or bacterial/viral infections based on white blood cell levels.

Biochemistry: Measures levels of calcium, magnesium, and phosphorus to detect deficiencies, as well as assesses organ damage.

Rapid Diagnostic Kits: Used for rapid detection of specific diseases (e.g., ELISA tests for antibody/antigen detection).

BOVINE DISEASES

Common Cattle Diseases Table

Disease	Cause	Key Symptoms	Transmission	Prevention/Control
Foot & Mouth (FMD)	Aphthovirus	Fever, blisters on mouth/hooves, lameness, drooling	Aerosol, direct contact, contaminated feed	Biannual vaccination, movement control
Bovine Respiratory Disease (BRD)	Viruses (IBR, BVD), bacteria (Pasteurella)	Coughing, fever, nasal discharge, labored breathing	Stress, poor ventilation, close contact	Vaccination, reducing stress, ventilation

Mastitis	Bacteria/fungi (e.g., *Staph aureus*)	Swollen/hard udders, clotted/off-color milk	Contaminated equipment, dirty bedding	Proper milking hygiene, antibiotic therapy
Bovine Viral Diarrhea (BVD)	BVD Virus (BVDV)	Diarrhea, weight loss, fever, infertility	PI (persistently infected) animals, nasal secretions	Vaccination, removing PI animals
Lumpy Skin Disease (LSD)	Capripoxvirus	Firm skin nodules, fever, swollen lymph nodes	Biting insects (flies, mosquitoes)	Vaccination, vector control
Anthrax	*Bacillus anthracis* (bacteria)	Sudden death, black blood from orifices	Ingestion of spores from soil	Annual vaccination
East Coast Fever (ECF)	*Theileria parva* (protozoa)	High fever, enlarged lymph nodes, death	Brown ear tick	Acaricide tick control, vaccination
Trypanosomiasis (Nagana)	*Trypanosoma* spp. (parasite)	Chronic wasting, fever, anemia	Tsetse flies	Vector control (traps), trypanocides
Brucellosis	*Brucella abortus* (bacteria)	Abortion (late pregnancy), retained placenta	Contaminated feed, aborted fetus	Vaccination, testing/segregation
Blackleg	*Clostridium chauvoei* (bacteria)	Sudden death, lameness, severe muscle swelling	Ingestion of spores	Vaccination

Nutritional Diseases

- **Milk Fever (Parturient Paresis):** Low blood calcium after calving, causing weakness and paralysis. *Control:* Proper dietary management in late pregnancy (DCAD diet).

- **Ketosis (Acetonemia):** Occurs in early lactation when energy intake does not meet demand, resulting in weight loss and drop in milk production. *Control:* Ensure high-energy intake postpartum.
- **Bloat:** Accumulation of gas in the rumen (frothy or free gas), leading to severe abdominal distension and breathing difficulty. *Control:* Gradual introduction to green pasture and anti-foaming agents.
- **Rickets/Osteomalacia:** Lack of calcium, phosphorus, or Vitamin D, causing weak, malformed bones.
-

Common Prevention & Control Measures

- **Vaccination:** Regular vaccination schedules for diseases like FMD, BVD, and Blackleg are the primary defense.
- **Biosecurity:** Quarantine new animals, limit visitors, and restrict animal movement.
- **Vector Control:** Use acaricides (dips/sprays) to kill ticks and control biting flies.
- **Hygiene & Sanitation:** Clean sheds daily, disinfect water troughs, and ensure proper waste disposal.
- **Nutritional Support:** Ensure a balanced diet to strengthen the immune system.
- **Key Symptoms of Illness**

  Look for a sudden drop in milk production, loss of appetite, changes in manure consistency, coughing, labored breathing, lameness, or abnormal skin nodules.

BY DR. MARTIN CHINEDU OBETTA.

SHEEP AND GOAT DISEASES

This guide outlines common, economically significant diseases affecting sheep and goats, categorized by their cause, along with management strategies to prevent and control them.

I. Viral Diseases

- **Peste des Petits Ruminants (PPR) / Goat Plague**
    - **Cause:** Morbillivirus (Paramyxoviridae family).
    - **Symptoms:** High fever, ocular/nasal discharge, mouth lesions, necrotic stomatitis, pneumonia, green diarrhea, high mortality in goats.
    - **Transmission:** Direct contact, inhalation, ingestion of contaminated water/feed.
    - **Prevention & Control:** Yearly vaccination, separating infected animals, quarantine of new animals.
- **Contagious Ecthyma / Orf / Scabby Mouth**
    - **Cause:** Parapoxvirus.
    - **Symptoms:** Thick, raised, scabby sores on lips, mouth, muzzle, udder, feet (lameness).
    - **Transmission:** Direct contact, scabs in the environment.
    - **Prevention & Control:** Vaccination, isolate infected, apply iodine/antimicrobial gels on scabs.
- **Bluetongue**
    - **Cause:** Orbivirus.
    - **Symptoms:** Fever, edema of face/lips/tongue, oral ulcers, lameness, abortion.
    - **Transmission:** Biting insects (*Culicoides* midges).
    - **Prevention & Control:** Vaccination, vector control (insecticides), avoiding grazing near water at dawn/dusk.
- **Sheep and Goat Pox**
    - **Cause:** Capripoxvirus.
    - **Symptoms:** Fever, pox lesions on skin (non-hairy parts), pustules, labored breathing.
    - **Transmission:** Contact, aerosols, scabs.
    - **Prevention & Control:** Vaccination, quarantine.

II. Bacterial Diseases

- **Enterotoxemia / Pulpy Kidney / Overeating Disease**
  - **Cause:** *Clostridium perfringens* Type C or D.
  - **Symptoms:** Sudden death, convulsions, diarrhea, often in fast-growing kids/lambs after diet change.
  - **Transmission:** Rapid multiplication of bacteria normally present in the gut.
  - **Prevention & Control:** Vaccination with CD&T, slow introduction to grain/lush pasture.
- **Caseous Lymphadenitis (CLA) / Abscesses**
  - **Cause:** *Corynebacterium pseudotuberculosis*.
  - **Symptoms:** Abscesses in lymph nodes (neck, jaw) or internal organs, weight loss.
  - **Transmission:** Discharge from ruptured abscesses, contaminated shears/equipment.
  - **Prevention & Control:** Culling/isolating infected animals, disinfecting equipment, shearing management.
- **Foot Rot**
  - **Cause:** *Dichelobacter nodosus* and *Fusobacterium necrophorum*.
  - **Symptoms:** Severe lameness, foul-smelling necrotizing infection between phooves.
  - **Transmission:** Contaminated, wet, muddy ground.
  - **Prevention & Control:** Hoof trimming, zinc sulfate/formalin foot baths, keeping areas dry.
- **Anthrax**
  - **Cause:** *Bacillus anthracis*.
  - **Symptoms:** Sudden death, dark bleeding from orifices.
  - **Transmission:** Ingestion of spores from contaminated soil.
  - **Prevention & Control:** Annual vaccination, burying/burning carcass, DO NOT OPEN carcass.
- **Brucellosis**
  - **Cause:** *Brucella melitensis* (primarily in goats).
  - **Symptoms:** Abortions (late pregnancy), retained placenta, epididymitis in males.

- **Transmission:** Contaminated fluids, milk, vaginal discharge.
- **Prevention & Control:** Testing, culling positive animals, sanitation.
- **III. Parasitic Diseases**

- **Internal Parasites / Worms (e.g., Barber's Pole Worm)**
  - **Cause:** *Haemonchus contortus* (helminths).
  - **Symptoms:** Anemia (pale gums/eyelids), bottle jaw (edema), weight loss, diarrhea.
  - **Transmission:** Grazing on pasture contaminated with larvae.
  - **Prevention & Control:** Strategic deworming (FAMACHA method), rotational grazing, pasture management.
- **Coccidiosis**
  - **Cause:** *Eimeria* protozoa.
  - **Symptoms:** Watery or bloody diarrhea, dehydration, weight loss (often young animals).
  - **Transmission:** Fecal-oral, overcrowding, wet bedding.
  - **Prevention & Control:** Clean water/housing, anticoccidials in fee

IV. Metabolic and Nutritional Diseases

- **Pregnancy Toxemia / Ketosis**
  - **Cause:** Energy imbalance, twin lamb disease.
  - **Symptoms:** Lethargy, inability to stand, coma, in late pregnancy.
  - **Prevention & Control:** Good nutrition/grain supplementation in late pregnancy.
- **Milk Fever / Hypocalcemia**
  - **Cause:** Low blood calcium, often in high-producing dairy goats.
  - **Symptoms:** Muscle tremors, weakness, recumbency.
  - **Prevention & Control:** Nutritional management, calcium supplements. **V. General Prevention Checklist (Biosecurity)**

1. **Quarantine new animals** for 30 days.
2. **Vaccinate annually** against common local diseases (PPR, CD&T).
3. **Keep pens clean and dry** to minimize bacterial and parasite load.
4. **Manage pastures** by rotating to reduce worm exposure.
5. **Isolate sick animals** immediately.

SWINE DISEASES

Here are common swine diseases:

1. Viral Diseases

- **African Swine Fever (ASF)**
    - **Cause:** ASF Virus (Asfarviridae family).
    - **Symptoms:** High fever ($40-42^{\circ}C$), loss of appetite, hemorrhages in skin (ears/abdomen), sudden death (100% mortality in acute forms).
    - **Transmission:** Direct contact with infected pigs/wild boar, ingesting contaminated pork/swill, soft ticks, and contaminated equipment/clothing.
    - **Prevention/Control:** No vaccine available. Strict biosecurity: ban swill feeding, quarantine new pigs, control insects, and cull infected herds.
- **Porcine Reproductive and Respiratory Syndrome (PRRS)**
    - **Cause:** PRRS Virus (Arterivirus).
    - **Symptoms:** Breeding failures (late-term abortions, mummies, weak piglets), blue ears, pneumonia, and high mortality in young pigs.
    - **Transmission:** Direct contact, aerosol (short distances), semen (artificial insemination), and contaminated needles.
    - **Prevention/Control:** Vaccination, all-in/all-out systems, purchasing stock from PRRS-negative farms, and air filtration.
- **Classical Swine Fever (CSF / Hog Cholera)**
    - **Cause:** CSF Virus (Pestivirus).
    - **Symptoms:** Fever, purple skin discoloration, constipation followed by diarrhea, neurological signs.
    - **Transmission:** Direct contact with secretions, feces, or blood; ingestion of contaminated garbage/pork.
    - **Prevention/Control:** Vaccination (in endemic areas), stamping-out policy (culling), and movement control.
- **Foot-and-Mouth Disease (FMD)**
    - **Cause:** Foot-and-mouth disease virus (Picornaviridae).

  - **Symptoms:** Blisters/vesicles on snout, hooves, and tongue; high fever, drooling, lameness.
  - **Transmission:** Highly contagious; direct contact, aerosols, and contaminated materials.
  - **Prevention/Control:** Vaccination, restricted movement, disinfection of vehicles, and prompt reporting.
- **Swine Influenza (Swine Flu)**
  - **Cause:** Influenza A virus (H1N1, H3N2).
  - **Symptoms:** Coughing, sneezing, fever, lethargy, difficulty breathing, decreased appetite.
  - **Transmission:** Rapidly spread via coughing/sneezing, and direct contact.
  - **Prevention/Control:** Annual vaccination, isolating sick pigs, and improved ventilation.
  - **2. Bacterial Diseases**
- **Erysipelas (Diamond Skin Disease)**
  - **Cause:** *Erysipelothrix rhusiopathiae*.
  - **Symptoms:** Diamond-shaped skin patches, high fever, joint pain/arthritis, sudden death.
  - **Transmission:** Feces and oronasal secretions of infected animals.
  - **Prevention/Control:** Vaccination and penicillin treatment.
- **Mycoplasma Pneumonia (Enzootic Pneumonia)**
  - **Cause:** *Mycoplasma hyopneumoniae*.
  - **Symptoms:** Persistent dry cough, slow growth, poor feed conversion, labored breathing.
  - **Transmission:** Direct contact (nose-to-nose) and aerosol.
  - **Prevention/Control:** Vaccination, better ventilation, and reducing overcrowding.
- **Swine Dysentery**
  - **Cause:** *Brachyspira hyodysenteriae*.
  - **Symptoms:** Severe bloody diarrhea with mucus, weight loss, and weakness.
  - **Transmission:** Fecal-oral route, rodents, and contaminated pens.
  - **Prevention/Control:** Rodent control, strict sanitation, and antibiotics (tiamulin, lincomycin).

- **Colibacillosis (E. coli)**
  - **Cause:** *Escherichia coli*.
  - **Symptoms:** Watery diarrhea in piglets (scours), dehydration, high mortality.
  - **Transmission:** Fecal-oral route.
  - **Prevention/Control:** Sanitation, keeping pens dry, and vaccination of sows.

3. Parasitic Diseases

- **Sarcoptic Mange**
  - **Cause:** *Sarcoptes scabiei var. suis* (mites).
  - **Symptoms:** Severe itching, skin lesions, hair loss, thickened skin, reduced weight gain.
  - **Transmission:** Direct contact between pigs.
  - **Prevention/Control:** Deworming (ivermectin), washing pigs with disinfectant, and isolating new pigs.
- **Ascariasis (Roundworms)**
  - **Cause:** *Ascaris suum*.
  - **Symptoms:** Weight loss, rough coat, reduced growth rates, "thumps" (respiratory distress).
  - **Transmission:** Ingestion of eggs from feces.
  - **Prevention/Control:** Regular deworming, daily cleaning of pens, and avoiding feeding on the ground.
- **Nutritional Disease:**

Causes of Piglet Anaemia

  - **Low Iron Stores at Birth:** Piglets are born with very low iron reserves (approx. 50mg), which are insufficient to support their rapid growth.
  - **Low Iron in Sow's Milk:** Sow's milk provides only a tiny amount of iron (about 1–2 mg/day), while piglets require 7-16 mg/day to sustain their growth.
  - **Indoor Housing/Concrete Floors:** Modern indoor farming prevents piglets from rooting in the soil, which is their natural source of iron.
  - **Rapid Growth Rates:** Fast-growing piglets require more iron to build blood, depleting their stores quicker.

- **Other Causes:** Occasionally, navel bleeding, vitamin K deficiency, or chronic blood loss from parasites can lead to anaemia.

Symptoms (Clinical Signs)

- Symptoms typically appear from 7 days onwards, with the risk period being the first 3-4 weeks of life:
- **Pale Skin and Mucous Membranes:** Noticeable pallor in the ears, nose, and gums.
- **"Thumps" (Laboured Breathing):** Rapid, shallow breathing, especially after exercise, as the piglet struggles for oxygen.
- **Listlessness/Lethargy:** Reduced activity, weakness, and reluctance to move.
- **Reduced Growth Rates:** Rough hair coat, wrinkled skin, and poor weight gain.
- **Scouring (Diarrhea):** Increased susceptibility to infections.
- **Sudden Death:** Severe cases can die suddenly due to heart failure.

- **Risk Period:** The most critical period is from 3 days old until the piglets begin to eat enough iron-fortified creep feed.
- **Impact:** If untreated, it can cause up to 10% of pre-weaning deaths.

Prevention and Treatment

Preventing iron deficiency is essential through iron supplementation:]

- **Iron Injections (Most Common & Effective):** Administer 150-200 mg of iron dextran or gleptoferron via intramuscular injection between 3-5 days of age. Injecting into the neck is preferred over the ham (leg) to reduce muscle damage/staining.
- **Oral Iron Supplements:** Iron pastes or liquids can be given, but these are less reliable and must be given on 2-3 occasions (days 7, 10, and 15).
- **Soil Access:** Allowing piglets access to clean, non-parasitic soil can provide natural iron.
    - **Creep Feed:** Providing iron-fortified creep feed early (after 7-10 days) helps, but does not replace the need for an initial injection

General Prevention and Control Strategies

- **Biosecurity:** Control visitor access, disinfect vehicles, and sanitize equipment.
- **Quarantine:** Isolate new pigs for 14–30 days before introduction.
- **Sanitation:** Regular cleaning of pens, proper waste management, and controlling pests (rodents, flies).

- **All-in/All-out Management:** Move groups of pigs of the same age together to break the disease cycle.
- **Vaccination Programs:** Tailor vaccination schedules to regional risks under veterinary guidance.
- **Good and adequate nutrition.**

POULTRY DISEASES

Common poultry diseases, their causes, symptoms, and control measures:

1. Viral Diseases

- **Newcastle Disease (NCD / Ranikhet)**
    - **Cause:** Paramyxovirus type 1.
    - **Symptoms:** Respiratory distress (coughing, sneezing), greenish diarrhea, nervous signs (twisted necks, circling), high mortality.
    - **Transmission:** Droppings, respiratory discharges, contaminated equipment, and feed.
    - **Prevention/Control:** Vaccination of chicks early (day 1-4) and regular boosters, quarantine new birds for 30 days.
- **Avian Influenza (Bird Flu / HPAI)**
    - **Cause:** Influenza Type A Virus.
    - **Symptoms:** Sudden death, lack of energy, swelling of head/comb/wattles, purple discoloration, respiratory distress.
    - **Transmission:** Wild birds, contaminated feed/water, and direct contact with infected birds.
    - **Prevention/Control:** Strict biosecurity, keep poultry indoors, control pests, and cull infected flocks.
- **Infectious Bursal Disease (Gumboro)**
    - **Cause:** Birna virus (highly resistant).

- **Symptoms:** Watery diarrhea, depression, ruffled feathers, sudden death in young chicks (3-6 weeks).
- **Transmission:** Fecal-oral, contaminated litter, feed, and water.
- **Prevention/Control:** Vaccination of breeders and young chicks, cleaning/disinfecting sheds.

- **Marek's Disease**
  - **Cause:** Herpes virus (affects nerves).
  - **Symptoms:** Paralysis of legs/wings, tumors, weight loss.
  - **Transmission:** Inhalation of dander and feather dust.
  - **Prevention/Control:** Vaccination of day-old chicks is the only effective control.
- **Fowl Pox**
  - **Cause:** Pox virus.
  - **Symptoms:** Warty growths on the comb/wattles, diphtheritic lesions in the mouth.
  - **Transmission:** Mosquitoes, lice, or direct contact.
  - **Prevention/Control:** Vaccination, mosquito control, and sanitation.

2. Bacterial Diseases

- **Salmonellosis (Pullorum Disease / Fowl Typhoid)**
  - **Cause:** *Salmonella pullorum* or *Salmonella gallinarum*.
  - **Symptoms:** White diarrhea, high mortality in chicks, weakness, huddled appearance.
  - **Transmission:** Vertical (hen to egg), contaminated environment.
  - **Prevention/Control:** Use certified disease-free hatcheries, test and slaughter carriers, proper sanitation.
- **Infectious Coryza**
  - **Cause:** *Haemophilus paragallinarum* (bacteria).
  - **Symptoms:** Swollen face/eyes, foul-smelling nasal discharge, reduced egg production.
  - **Transmission:** Direct contact, contaminated water.
  - **Prevention/Control:** Good biosecurity, culling carriers, vaccination.
- **Colibacillosis**
  - **Cause:** *Escherichia coli* (E. coli).
  - **Symptoms:** Listlessness, ruffled feathers, labored breathing, yellow-green diarrhea.
  - **Transmission:** Fecal contamination, contaminated litter.

  - **Prevention/Control:** Good litter management, ventilation, and clean water.
- **Blackhead Disease (Histomoniasis)- mainly turkey.**
  - **Cause:** A protozoan parasite (*Histomonas meleagridis*) that attacks the liver and caeca.
  - **Transmission:** Ingesting cecal worm eggs (*Heterakis gallinarum*) contaminated with the parasite, often carried by chickens (which are resistant) or earthworms.
  - **Symptoms:** Lethargy, ruffled feathers, drooping wings, and bright yellow diarrhoea.
  - **Prevention:** Do not house turkeys with chickens or on ground used by chickens within the last 10 years. Regular deworming to control cecal worms is essential.
- **Mycoplasmosis (Respiratory Disease/Sinusitis)**
  - **Cause:** *Mycoplasma gallisepticum* or *M. synoviae* bacteria.
  - **Transmission:** Direct contact with infected birds, aerosolized droplets, or transmission from hen to chick through the egg.
  - **Symptoms:** Coughing, sneezing, snicking (sneezing sound), swollen sinuses/face, and joint swelling leading to lameness.
  - **Prevention:** Purchase poults from certified disease-free breeders, maintain good ventilation, and reduce stress.
- **Fowl Cholera**
  - **Cause:** Bacteria (*Pasteurella multocida*).
  - **Transmission:** Contaminated water, equipment, wild birds, and rodents.
  - **Symptoms:** Sudden death, greenish diarrhea, lameness, and swollen wattles/face.
- **Prevention:** High sanitation standards, prompt disposal of carcasses, and vaccination.

3. Parasitic & Other Diseases

- **Coccidiosis**
  - **Cause:** *Eimeria* protozoa.
  - **Symptoms:** Bloody diarrhea, yellowish droppings, ruffled feathers, weight loss.
  - **Transmission:** Ingestion of oocysts from wet, contaminated litter.

  - **Prevention/Control:** Keep litter dry, use anticoccidial drugs (coccidiostats) in feed.
- **Worm Infestations (Roundworms/Tapeworms)**
  - **Cause:** Internal parasites.
  - **Symptoms:** Weight loss, poor growth, pale combs, diarrhea.
  - **Transmission:** Droppings, contaminated food/water.

- **Prevention/Control:** Regular deworming.
- **General Prevention and Control Strategies**

- **Common Avian Ectoparasites**
    - **Red Mites (Dermanyssus gallinae):** Known as roost mites, they live in the housing and feed on blood at night.
    - **Northern Fowl Mites (Ornithonyssus sylviarum):** Spend their entire life cycle on the bird, usually around the vent.
    - **Lice (Mallophaga):** Feed on feathers and skin debris, causing irritation.
    - **Ticks (Argas persicus):** Feed on blood, causing anemia.
    - **Fleas (e.g., Sticktight flea):** Attach firmly to the skin, causing ulcers.
    - **Flies (Hippoboscid flies):** Blood-feeding flies, often seen in wild birds. **Causes of Infestation**
    - **Direct Contact:** Transmission between birds, particularly in gregarious species.
    - **Environmental Contamination:** Mites can hide in cracks, crevices, and nesting materials, surviving for months without a host.
    - **Wild Bird Interaction:** Wild birds often bring parasites to backyard poultry.
    - **Poor Husbandry:** Dirty, crowded, or damp environments facilitate the rapid spread of parasites.]

Transmission and Impact

Ectoparasites cause damage directly through feeding and indirectly by transmitting diseases:

- **Disease Vectors:** Ticks and mites can transmit fowl cholera, *Pasteurella multocida*, *Aegyptianella pullorum*, and viral diseases like Newcastle disease.
- **Physical Damage:** Intense irritation, feather damage, dermatitis, anemia, and significant weight loss or reduced egg production.
- **Immune Suppression:** Severe infestations can lead to high mortality, particularly in young birds.
- **Human Health:** Some mites (e.g., Red Mite) can feed on humans, causing irritation.

Prevention and Control

Effective management requires a combination of treatment for the birds and the environment.

GENERAL PREVENTION AND CONTROL

- **Biosecurity:** Limit visitors, use footbaths, quarantine new birds for 30 days.
- **Vaccination:** Adhere to a strict, region-specific vaccination schedule.
- **Hygiene:** Keep houses dry, clean, and well-ventilated to prevent bacterial/protozoal growth.
- **Pest Control:** Control rodents and insects, which act as disease vectors.
- **Isolation:** Separate sick birds immediately.

Note: Many viral diseases have no treatment; prevention through vaccination and biosecurity is critical.

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MICROLIVESTOCK DISEASES

Microlivestock, including rabbits, guinea pigs, and poultry (such as quail), are prone to various infectious and metabolic diseases that can devastate small-scale farming operations if not managed. Key preventative measures for all species include biosecurity, high sanitation standards, quarantine of new animals, and stress reduction.

Below is a list of common microlivestock diseases categorized by species.

1. Rabbit Diseases

- **Snuffles (Pasteurellosis)**
    - **Cause:** *Pasteurella multocida* bacterium.
    - **Symptoms:** Runny nose/eyes, sneezing, matted fur on front paws, head tilt, pneumonia.
    - **Transmission:** Direct contact with infected rabbits, aerosolized sneeze particles.
    - **Prevention/Control:** Isolate infected rabbits, improve ventilation, reduce stress, antibiotic treatment.
- **Coccidiosis**
    - **Cause:** Protozoa (*Eimeria* species).
    - **Symptoms:** Diarrhea, blood-stained feces, pot belly, weight loss, death, especially in young rabbits.
    - **Transmission:** Ingestion of contaminated feed or water with feces.
    - **Prevention/Control:** Keep cages dry and clean, use wire-bottomed cages, use anticoccidial drugs in water.
- **Myxomatosis**
    - **Cause:** Myxoma virus (poxvirus).
    - **Symptoms:** Swollen head, eyes, and genitals; thick eye discharge, high fever, sudden death.
    - **Transmission:** Insects (mosquitoes, fleas) or direct contact.
    - **Prevention/Control:** No treatment; use mosquito-proof hutch, flea control, vaccination.
- **Rabbit Hemorrhagic Disease (RHDV2/Calicivirus)**
    - **Cause:** Highly contagious Calicivirus.
    - **Symptoms:** Fever, lethargy, sudden death within 24-72 hours, bleeding from nostrils.
    - **Transmission:** Direct contact, indirect contact via fomites (clothing, shoes), insects.
    - **Prevention/Control:** Vaccination, strict biosecurity, quarantine of new animals.
- **Mange/Ear Canker (Ectoparasites)**
    - **Cause:** Mites (e.g., *Psoroptes cuniculi*).

   - **Symptoms:** Crusty brownish scabs in the ear, scratching, head shaking.
   - **Transmission:** Direct contact.
   - **Prevention/Control:** Topical or injectable medication (ivermectin), regular cleaning.

2. Guinea Pig Diseases

- **Respiratory Infection (Pneumonia)**
   - **Cause:** Bacteria (*Bordetella bronchiseptica*, *Streptococcus pneumoniae*).
   - **Symptoms:** Sneezing, coughing, runny eyes/nose, labored breathing, lethargy, weight loss.
   - **Transmission:** Direct contact, aerosol transmission, carriers (sometimes rabbits).
   - **Prevention/Control:** Avoid overcrowded conditions, keep cage clean, keep away from rabbits.
- **Bumblefoot (Pododermatitis)**
   - **Cause:** Bacterial infection (*Staphylococcus aureus*) caused by pressure.
   - **Symptoms:** Swollen, inflamed, or crusty footpads; lameness.
   - **Transmission:** Dirty, wet, or wire-bottomed cages.
   - **Prevention/Control:** Use soft bedding, clean cages frequently, and prevent obesity.
- **Scurvy (Vitamin C Deficiency)**
   - **Cause:** Dietary lack of Vitamin C (cannot produce their own).
   - **Symptoms:** Weakness, painful joints, reluctance to move, poor coat, bleeding gums.
   - **Prevention/Control:** Daily supplementation with fresh veggies (kale, peppers) or Vitamin C tablets, not through water.
- **Ringworm**
   - **Cause:** Fungal infection.
   - **Symptoms:** Patchy hair loss, dry/flaky skin, itching around face/ears.
   - **Transmission:** Direct contact, highly contagious to humans.
   - **Prevention/Control:** Isolate infected animal, disinfect, antifungal treatment.

3. Poultry (Quail/Backyard Chicken) Diseases

- **Newcastle Disease (ND)**
   - **Cause:** Paramyxovirus.
   - **Symptoms:** Greenish droppings, twisted necks, gasping, drop in egg production.
   - **Transmission:** Contaminated feed, water, aerosol from infected birds.

    - **Prevention/Control:** Vaccination, strict biosecurity, separating sick birds.
- **Coccidiosis**
    - **Cause:** Protozoa (*Eimeria* species).
    - **Symptoms:** Ruffled feathers, bloody diarrhea, pale combs, stunted growth.
    - **Transmission:** Feces-contaminated soil/litter.
    - **Prevention/Control:** Keep brooder dry, use medicated starter feed, good sanitation.
- **Fowl Pox**
    - **Cause:** Poxvirus.
    - **Symptoms:** Scabby lesions on comb/wattles, "wet form" lesions in the mouth.
    - **Transmission:** Mosquito bites, or scabs coming into contact with wounds.
    - **Prevention/Control:** Vaccination, insect control.
- **Salmonellosis**
    - **Cause:** *Salmonella* bacteria.
    - **Symptoms:** Diarrhea, weakness, drooping wings, sudden death.
    - **Transmission:** Contaminated feed/water, feces.
    - **Prevention/Control:** Good hygiene, purchasing clean stock

General Disease Control Principle

- **Quarantine:** Isolate new additions for at least 2 weeks.
- **Hygiene:** Clean cages, feeders, and waterers regularly.
- **Ventilation:** Prevent ammonia buildup in housing to reduce respiratory issues.
- **Nutrition:** Provide a balanced diet to support immune system health.

FISH DISEASES

Common fish diseases often stem from environmental stress (poor water quality, overcrowding) and pathogens like bacteria, viruses, parasites, or fungi. Common illnesses and their management include:

Ichthyophthiriasis (Ich or White Spot Disease) (Parasitic)

- **Causes:** *Ichthyophthirius multifiliis* (protozoan parasite).
- **Symptoms:** White spots resembling salt grains on skin, gills, and fins; rubbing against objects.
- **Transmission:** Direct parasite transfer between fish, especially in dense populations.

- **Prevention/Control:** Increase water temperature, quarantine new fish, use specialized chemical treatments (e.g., formaldehyde, malachite green).

- **Fin and Tail Rot (Bacterial)**
    - **Causes:** Bacteria such as *Aeromonas* or *Pseudomonas*.
    - **Symptoms:** Frayed, frayed, or deteriorating fins; red, inflamed fin base.
    - **Transmission:** Poor water conditions, injuries, stress.
    - **Prevention/Control:** Improve water quality, treat with antibiotics (e.g., tetracycline), ensure proper nutrition.
- **Dropsy (Bacterial/Internal Organ Failure)**
    - **Causes:** *Aeromonas* bacteria, typically in poor water environments.
    - **Symptoms:** Abdomen swollen with fluid, scales protruding (pinecone appearance).
    - **Transmission:** Contaminated water, feces.
    - **Prevention/Control:** Regular cleaning, proper feeding, isolating sick fish, antibiotics (if caught early).
- **Columnaris Disease (Bacterial)**
    - **Causes:** *Flavobacterium columnare*.
    - **Symptoms:** White, ragged lesions on the mouth or body, cotton-like patches.
    - **Transmission:** Stress, poor quality water.
    - **Prevention/Control:** Minimize handling, maintain lower temperatures if possible, use antibiotics.
- **Parasitic Crustaceans (e.g., Anchor Worm)**
    - **Causes:** *Lernaea* species (crustacean).
    - **Symptoms:** Visible worm-like parasites attached to the body, causing ulcers.
    - **Transmission:** Introduction of infested fish or plants.
    - **Prevention/Control:** Quarantine new arrivals, physical removal with tweezers, specific antiparasitic medicines.
    - **Common Causes of Disease Outbreaks**

1. **Poor Water Quality:** High ammonia, nitrites, or low oxygen.
2. **Stress:** Caused by poor handling, transport, or inappropriate water temperature.
3. **Overcrowding:** Increases stress and speeds up transmission.
4. **Malnutrition:** Weakens the immune system.

General Prevention and Control Measures

- **Quarantine:** Isolate new fish for at least 3-4 weeks.
- **Water Management:** Maintain regular partial water changes, ensure proper filtration, and monitor pH and chemical levels.
- **Nutrition:** Feed a high-quality, balanced diet.
- **Hygiene:** Sanitize nets and equipment between tanks to avoid spreading pathogens

TYPES AND LIFE CYCLE OF HELMINTHES AND ECTO PARASITES OF LIVESTOCK

Helminths (internal parasitic worms) and ectoparasites (external parasites) are significant causes of morbidity and reduced productivity in livestock, including cattle, sheep, goats, pigs, and poultry. Their life cycles, which can be direct (single host) or indirect (requiring intermediate hosts), are critical to understand for effective management and control.

Helminths (Internal Parasites)

Helminths are divided into three main groups based on structure: nematodes (roundworms), trematodes (flukes), and cestodes (tapeworms).

1. Nematodes (Roundworms

These are the most prevalent internal parasites in livestock, inhabiting the intestinal lumen and extraintestinal sites.

- **Types:** *Haemonchus* (wireworm), *Ostertagia* (brown stomach worm), *Cooperia* (small intestinal worm), *Trichostrongylus* (black scour worm), *Dictyocaulus* (lungworm), and *Ascaris suum* (pig roundworm).
- **Life Cycle (Direct):** Adult worms in the animal (definitive host) lay eggs, which are passed in feces. The eggs hatch and develop into L3 larvae on the pasture (3–7 days). Livestock ingest the L3 larvae while grazing. The larvae then migrate to the gastrointestinal tract (e.g., abomasum) or, in the case of lungworms, migrate to the lungs, maturing into adults.
- **Special Considerations:** Some species (e.g., *Ostertagia*) can undergo hypobiosis (arrested development) within the host during hostile environmental conditions.

2. Trematodes (Flukes)

These are leaf-shaped flatworms requiring aquatic snails as intermediate hosts.

- **Types:** *Fasciola hepatica* (liver fluke) and *Paramphistomum* (rumen fluke).
- **Life Cycle (Indirect):** Eggs are passed in feces and must reach water. They hatch into miracidia, which penetrate a mud snail (*Lymnaea* species). Inside the snail, they develop into cercariae, which emerge and encyst as metacercariae on vegetation. Livestock ingest these cysts while grazing. The immature flukes migrate through the liver parenchyma to the bile ducts, maturing in 8–12 weeks.

3. Cestodes (Tapeworms)

These are segmented flatworms that inhabit the intestine.

- **Types:** *Moniezia* (ruminant tapeworm) and *Taenia* species.

- **Life Cycle (Indirect):** Gravid proglottids are passed in feces. Intermediate hosts (e.g., mites for *Moniezia*, pigs/cattle for *Taenia solium/saginata*) ingest the eggs. The larval stage (cysticercus) develops in the tissue of the intermediate host. The final host becomes infected by ingesting the larval stage.

ECTOPARASITES (EXTERNAL PARASITES)

Ectoparasites live gon the skin, in the fleece, or on the hair of livestock.

1. Ticks

- **Types:** *Boophilus microplus*, *Amblyomma variegatum*, and *Rhipicephalus* species.
- **Life Cycle:** Eggs are laid in the environment. Six-legged larvae hatch, crawl onto vegetation, and attach to a host. They feed, molt into eight-legged nymphs, and often drop to the ground to molt into adults, or remain on the host (one-host ticks).

2. Lice

- **Types:** Chewing lice (e.g., *Bovicola bovis*) and Sucking lice (e.g., *Linognathus vituli*).
- **Life Cycle:** Lice spend their entire life cycle on the host, passing through egg (nit), nymph, and adult stages. They are highly host-specific.

3. Mites (Mange)

- **Types:** *Sarcoptes scabei* (sarcoptic mange) and *Psoroptes ovis* (sheep scab).
- **Life Cycle:** Mites burrow into the skin, causing severe dermatitis. Their entire life cycle, including egg-laying, takes place on the host.

4. Flies and Myiasis (Flystrike)

- **Types:** Blowflies (*Lucilia* spp.), Warble flies (*Hypoderma* spp.), and Horn flies (*Haematobia* irritans).

- **Life Cycle:** Flies lay eggs in wounds or soiled wool (flystrike). The hatching larvae (maggots) feed on the animal's tissue. Some larvae, like warbles, migrate within the host before emerging to pupate in the soil.

Key Differences in Livestock Species

Livestock	Key Helminths	Key Ectoparasites
Cattle	*Ostertagia*, *Cooperia*, *Fasciola hepatica*	Ticks, Sucking/Biting lice, Horn flies, Mange

Sheep	*Haemonchus contortus*, *Teladorsagia*, *Fasciola*	Sheep scab, Keds, Blowflies, Ticks
Pigs	*Ascaris suum*, *Trichuris suis*	Sarcoptic mange, Hog lice (*Haematopinus suis*)
Poultry	*Ascaridia galli*, *Heterakis gallinarum*	Red mites (*Dermanyssus gallinae*), Lice

Management and Control

Integrated parasite management is crucial to minimize reliance on chemicals, as anthelmintic and acaricide resistance is increasing.

- **Rotational Grazing:** Moving animals to clean pastures prevents ingestion of larvae.
- **Targeted Treatment:** Using faecal egg counts (FEC) to treat only high-shedding animals reduces selection pressure for resistance.
- **Habitat Control:** Draining wetland areas reduces snail intermediate hosts for liver flukes.
- **Housing Hygiene:** Cleaning pens removes ticks and lice

HOST PARASITES RELATIONSHIP IN LIVESTOCK

The host-parasite relationship in livestock is a complex, often long-term association where parasites (helminths, protozoa, or arthropods) exploit hosts for nourishment and survival. This relationship causes economic losses through poor growth, reduced production, infertility, or death, with the host often mounting immune defenses while the parasite evolves to evade them.

Key Aspects of the Relationship:

- **Parasitism Definition:** Parasites live on (ectoparasites) or inside (endoparasites) a host, causing harm while obtaining nutrients without immediately killing the animal.
- **Immune Response vs. Evasion:** Livestock (e.g., cattle, sheep) defend themselves through innate and adaptive immunity, such as creating inflammation, while parasites like *Trypanosoma* evade control by changing surface proteins.

- **Pathology and Impact:** Parasites cause significant harm by creating disorders in hematopoiesis (blood formation), suppressing the immune system, and damaging digestion. Examples include liver damage from *Dicrocoelium dendriticum* in sheep, respiratory failure from *Syngamus trachea* in poultry, and anemia from tsetse-transmitted trypanosomes in cattle.
- **Host-Parasite Dynamics:**
  - **Infection Dynamics:** The intensity of disease is often related to the quantity of parasites initially introduced (e.g., sporozoites in *Theileria parva*).
  - **Immune Tolerance:** Some indigenous livestock, like N'Dama cattle, display "trypanotolerance," enabling them to control parasitaemia better than susceptible breeds.
  - **Environmental Influence:** Free-range conditions increase risks, as animals interact more with intermediate hosts (slugs, snails) or contaminated grazing areas.
- **Management Strategies:** Effective control requires understanding the parasite life cycle, such as targeting the larval stages or limiting exposure to contaminated grazing, as highlighted in this CCE flock talk regarding internal and external parasite management.

Key Parasite Types:

- **Internal (Helminths/Nematodes):** Intestinal worms impacting poultry and ruminants.
- **External (Ectoparasites):** Fleas or ticks that feed on blood.
- **Protozoa:** Single-celled organisms such as *Trypanosoma* that cause chronic diseases in cattle.

BY DR. MARTIN CHINEDU OBETTA.

EFFECTS OF ECTOPARASITES ON LIVESTOCK PRODUCTION

Ectoparasites—including ticks, lice, mites, and flies—severely impact livestock production by causing significant weight loss, reduced milk yield, hide damage, and transmitting diseases. They cause blood loss, skin irritation, and stress, leading to decreased feed efficiency and increased mortality, particularly in young or undernourished animals. These parasites often result in major economic losses through reduced productivity and high control costs.

Key Effects of Ectoparasites on Livestock

- **Production Losses (Meat and Milk):** Heavy infestations decrease weight gain, reduce meat quality, and cause substantial declines in milk yield. For example, tick-infested dairy cows can lose 90.24 liters of milk per lactation in Brazil, and in Argentina, tick-transmitted diseases can cause up to 38% mortality.
- **Disease Transmission:** Ticks and flies are major vectors for diseases, including babesiosis, anaplasmosis, and heartwater, which can lead to high fever, severe illness, or death in livestock.

- **Hide and Wool Damage:** Mites and lice cause skin diseases, such as mange, which results in severe pruritus (itching), hair loss, and scarring, lowering the value of hides. Sheep keds and lice cause wool loss, staining, and damage to the skin.
- **Physiological Stress and Nutritional Impact:** Parasites cause blood loss and high skin irritation, which changes the animal's energy balance and reduces feeding time due to restlessness.
- **Economic Impact:** Beyond direct losses, producers face costs for acaricides and insecticides, increased labor for treatment, and potential restrictions on animal movement.

Common Ectoparasites and Their Damages

- **Ticks:** Cause blood loss, damage hides, and transmit diseases like babesiosis.
- **Lice:** Sucking lice (e.g., *Linognathus vituli*) cause anemia, while chewing lice (e.g., *Bovicola bovis*) cause irritation, hair loss, and damage to hides.
- **Flies:** Stable flies (*Stomoxys calcitrans*) and horn flies create painful bites, reducing milk and meat production.
- **Mites:** Cause mange, leading to severe skin damage and reduced animal value.

GENERAL MEASURES FOR DISEASE PREVENTION IN ANIMALS

Animal disease prevention relies on a combination of strict biosecurity, routine vaccination, proper nutrition, and environmental hygiene.

Key measures include quarantining new or sick animals, controlling access to facilities, implementing parasite control (deworming/dipping), and regular veterinary surveillance to ensure early detection. [

Prevention Strategies

- **Biosecurity Measures:** Control the movement of people, equipment, and vehicles entering animal areas to prevent pathogen introduction. Implement disinfection protocols and provide protective clothing.
- **Vaccination Programs:** Maintain up-to-date vaccinations for livestock and pets to build herd immunity and protect individuals against common pathogens.
- **Quarantine and Isolation:** Isolate new arrivals for at least 10–14 days and immediately segregate sick animals to prevent the spread of infection.

- **Environmental Hygiene:** Regularly clean and disinfect housing, maintain clean water supplies, and ensure proper waste disposal.
- **Nutritional Management:** Provide a balanced diet to strengthen the immune system and increase resistance to diseases.
- **Parasite Control:** Use deworming medications and pesticides (dipping/spraying) to control internal and external parasites.
- **Veterinary Surveillance:** Conduct regular health checks and work with professionals to establish treatment protocols, as outlined on the WOAH Animal Health guidelines. **Specific Protective Actions**
- **Rodent and Pest Control:** Keep food storage secure and eliminate pests that act as disease vectors.
- **Safe Feed and Water:** Ensure water is potable and feed is stored properly to prevent contamination.
- **Worker Education:** Train staff on hygiene practices, including proper handwashing, and ensure they use personal protective equipment.

IMPORTANCE OF VETERINARY SERVICES ON LIVESTOCK PRODUCTION

Veterinary services are essential for maximizing livestock production and must be included starting from the planning period of the farm (i.e. before siting commences).

Below are a few of the very key contributions of Veterinarians in livestock poduction:

- **Disease Prevention & Control:** Veterinarians perform routine vaccinations, implement parasitic control programs, and manage outbreak control to prevent disease spread and reduce livestock mortality.
- **Improvement of Production:** Through expert consultancy, nutrition advice, and breeding support (e.g., artificial insemination), vets help increase meat, milk, and wool production.
- **Food Safety and Public Health:** Veterinarians oversee farm-level hygiene and carry out pre-slaughter inspection to guarantee the safety of animal products, preventing diseases like tuberculosis and anthrax from entering the food chain.

- **Advisory Services for Farmers:** Vets educate farmers on optimal modern livestock management techniques, which increases awareness and leads to more sustainable agricultural practices.
- **Drug Management & Residue Prevention:** They ensure proper use of antibiotics and medicines to treat animals while ensuring that drug residues in meat and milk do not exceed safety standards.

Without proper veterinary services, farmers are at risk of significant economic losses due to high disease incidence and low herd productivity.

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